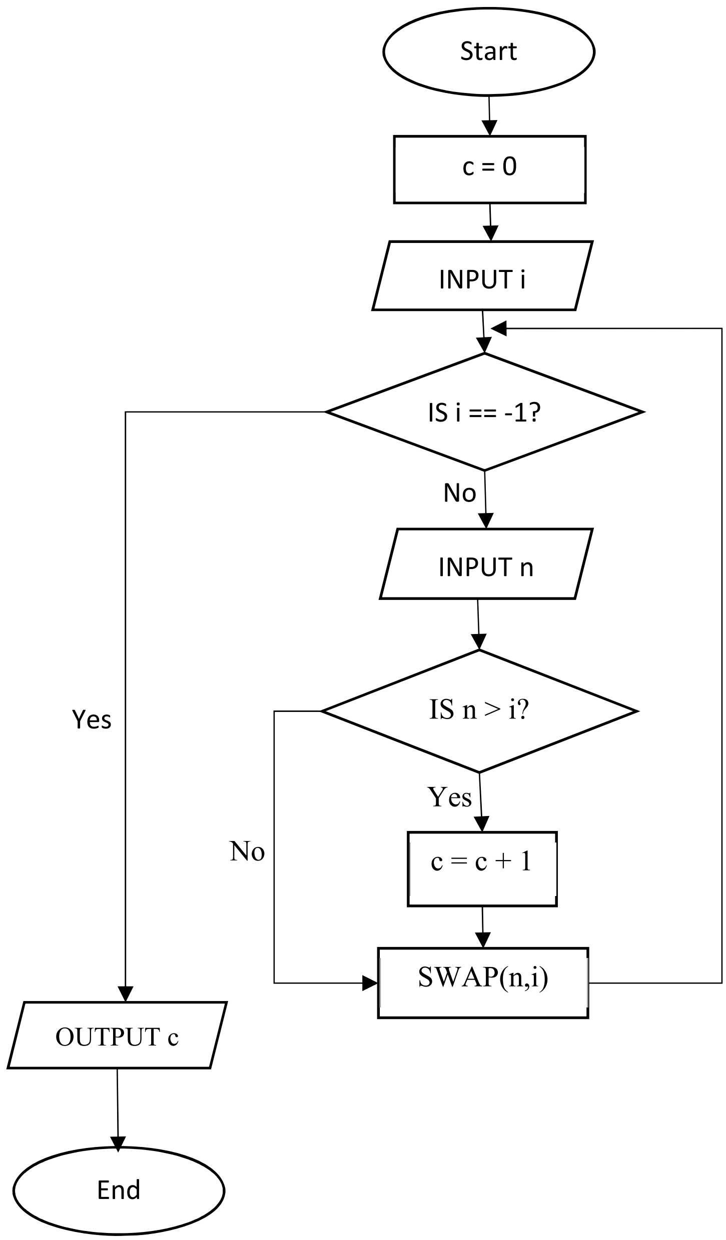
Qn	Ans
Q1 (a)	Start c = 0 INPUT i IS i == -1? No Yes INPUT n IS n > i? Yes No c = c + 1 SWAP(n,i) OUTPUT c End

Q1 (b)

c	i	n	OUTPUT
0			
0	72		
0	72	69	
0	69	72	
1	69	76	
1	69	76	
1	76	69	
1	76	76	
1	76	76	
1	76	79	
2	79	76	
2	79	-1	
2	-1	79	2

c	i	n	output			is n>i
0						
0	72				input i [72]	
0	**72**	**69**			**if i<>-1 input n [69]**	
0	**72**	**69**			**is n>I**	
0	69	72			swap	
0	**69**	**76**			**if i<>-1 input n [76]**	
0	**69**	**76**			**is n>I**	**TRUE**
1	76	69			c++, swap	
1	**76**	**76**			**if i<>-1 input n [76]**	
1	**76**	**76**			**is n>I**	
1	76	76			swap	
1	**76**	**79**			**if i<>-1 input n [79]**	
2	79	76			c++, swap	
2	**79**	**-1**			**if i<>-1 input n [-1]**	
2	**79**	**-1**			**is n>I**	
2	-1	79			swap	
2	-1	79	2		if i<>-1 output c	TRUE

Q1(c)

The purpose of this operation is to find out how many letters in a word have larger ASCII value than the preceding letter.
OR

The purpose of this operation is to find the number of times that a letter was alphabetically greater than the preceding letter or the letter directly before it.

Q1 (d)(i)

$$79_{10} = 4F_{16}$$

Denary	Quotient	Remainder	Hexadecimal
79 / 16	4	15	4
15 /16	0	15	F

Q1 (d)(ii)

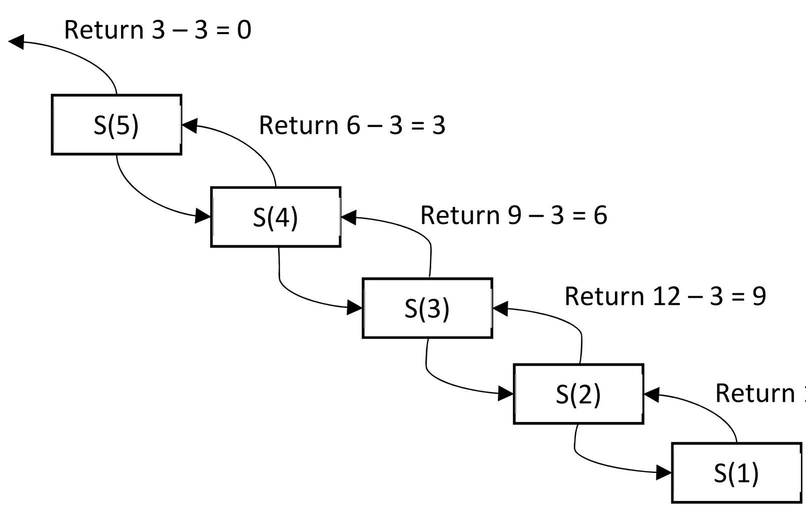
$$79_{10} = 0100\ 1111_2$$

Denary	Quotient	Remainder
79 / 2	39	1
39 / 2	19	1
19 / 2	9	1
9 / 2	4	1
4 / 2	2	0
2 / 2	1	0
1 / 2	0	1

Q1 (d) (iii)

Using ASCII as a method for encoding characters only limits the encoding to English characters. If used for characters of other languages, ASCII cannot be used.

Q2 (a)



Final return value is 0

Q2 (b)

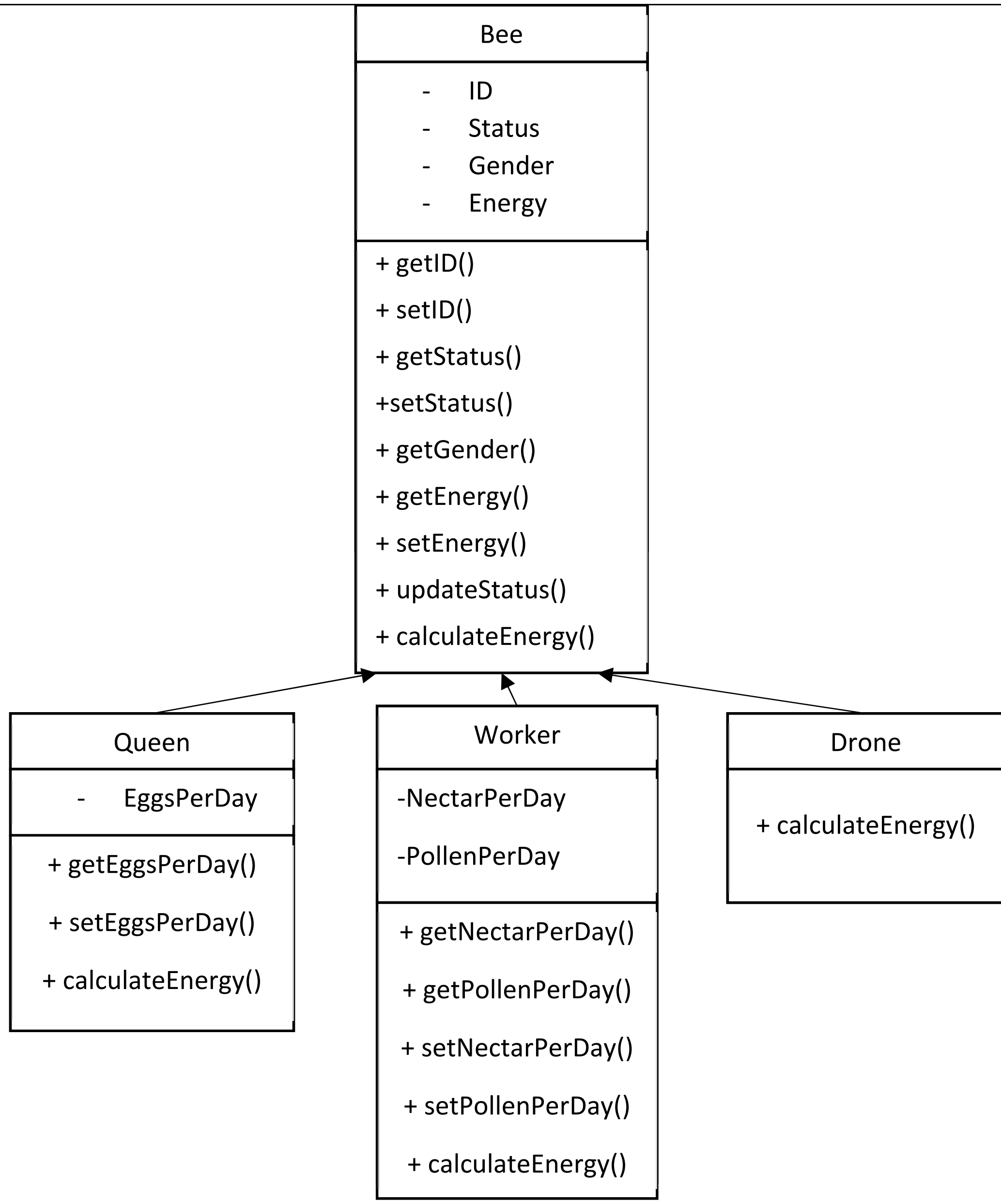
1. Line 05, where the function calls itself S(n-1), shows that it is a recursive function.
2. Line 02,03 showing the terminating condition of n = 1, also shows that is a recursive function.
3. Subsequence function call would converge on the base case

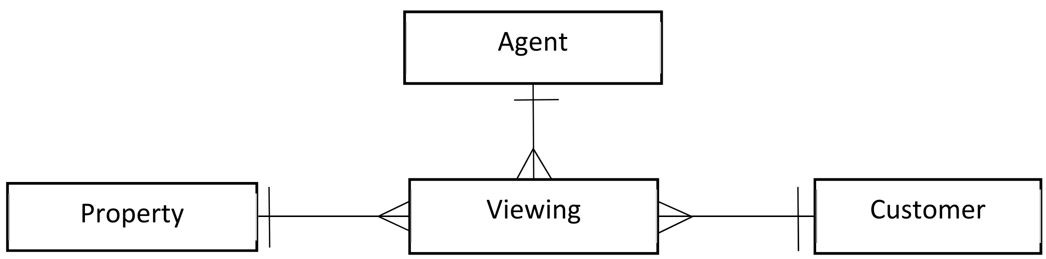
Q2 (c)

When a S(2) is called, a new activation record that holds the function's parameters, local variables and the return address, is pushed onto a run-time stack.

	When S(1) is called, a new activation record with the same type of data is pushed onto the stack. Since S(1) is the terminating case, it will start to return information. When the called function returns, its activation record is popped from the stack. The activation record is used to restore the environment of the interrupted function and resume execution of the interrupted function.
Q2 (d)	The recursion will continue indefinitely because the function does not reach its base case. As a result, the function keeps calling itself, eventually leading to a stack overflow runtime error when the system runs out of memory.

Q2 (e)	<pre>FUNCTION Iterative_S(n: INTEGER) RETURNS INTEGER
	    DEFINE result : INTEGER
	    result ← 12
	    WHILE n <> 1
	        result ← result – 3
	        n ← n – 1
	    ENDWHILE
	    RETURN result
	ENDFUNCTION</pre>

Q3(a)	
Q3 (b)	Instantiation is the process of creating an object from a class definition. The object created was an instance of a class.
Q3(c)(i)	Encapsulation is the bundling of data attributes and the methods that operate on them within a class. It restricts direct access to certain attributes from outside the class, thereby protecting the internal state of the object and exposing controlled access through public methods. For example, if a class Bee has a private attribute ID, code outside the class cannot access or modify ID directly. Instead, it must use public methods like getID() to retrieve the value or setID() to modify it.
Q3(c)(ii)	Inheritance is a feature in object oriented programming that allows a subclass to inherit attributes and methods from an existing class, known as superclass. For example, the Worker subclass can inherit attributes Status and Gender and methods setStatus(), getStatus(), setGender() and getGender() from the Bee superclass.

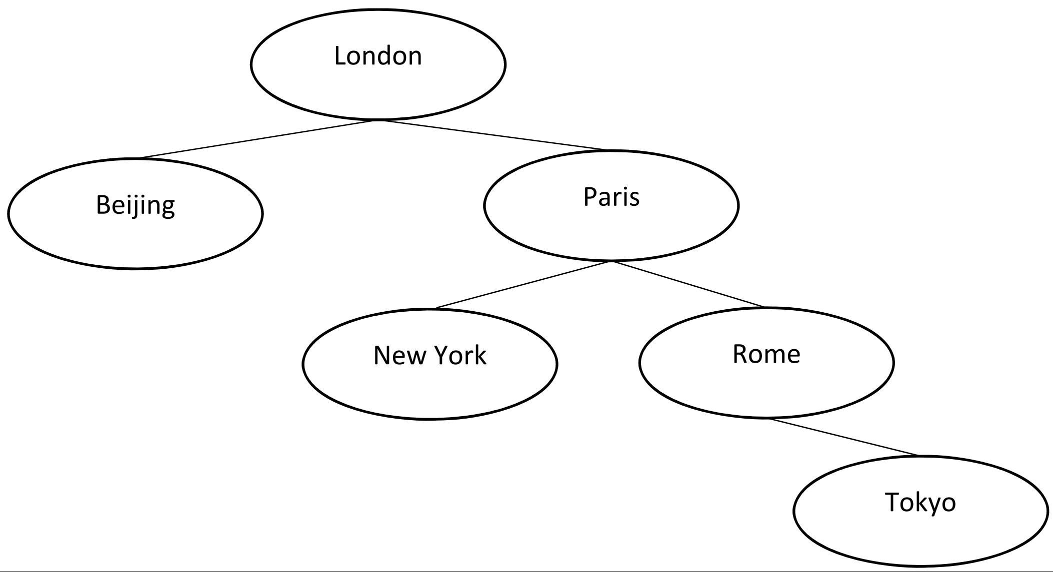
<table>
<tr><td>Q3(c)(iii)</td><td>Polymorphism is the object’s ability to take different forms. It allows the methods in the subclass to have the same name as the superclass but with different behavior. An example of this would be through the Queen method of calculateEnergy() , which override the calculateEnergy() of the Bee class, having a different algorithm compared to the superclass Bee, or the other subclasses Worker or Drone that have the same method name.</td></tr>
<tr><td>Q3(d)</td><td>
<table>
<tr><th>Type of test</th><th>Test data</th></tr>
<tr><td>Normal</td><td>15</td></tr>
<tr><td>Abnormal</td><td>40</td></tr>
<tr><td>Extreme</td><td>0</td></tr>
</table>
</td></tr>
<tr><td>Q4(a)(i)</td><td>


</td></tr>
<tr><td>Q4(a)(ii)</td><td>Property(PropertyReferenceCode, Address, Price, BedroomNo)

Agent(AgentReferenceNo, Name, Salary)

Customer(CustomerReferenceNumber, Name, PhoneNo, Email)

Viewing(PropertyReferenceCode, Date, Time, AgentReferenceNo, CustomerReferenceNo)</td></tr>
<tr><td>Q4 (a) (iii)</td><td>CREATE TABLE Property(
PropertyReferenceCode TEXT,
Address TEXT,
Price INTEGER,
BedroomNo INTEGER,
PRIMARY KEY (PropertyReferenceCode)
)</td></tr>
<tr><td>Q4 (b) (i)</td><td>Backing up data are needed to protect critical information from loss due to hardware failure, cyberattacks, accidental deletion, or disasters, and to ensure that important data can be quickly and accurately recovered to maintain business operations and minimise downtime.
OR
Backing up data are needed to ensure that important data, files, or information can be recovered in case of loss, corruption, or damage. Backups serve as a form of insurance against data loss and provide a way to restore critical data to its previous state.</td></tr>
</table>

<table>
<tr><td>Q4 (b)(ii)</td><td>The need of archiving is to preserve and protect information and records that are no longer actively in use but still have historical, legal, or cultural value. Archiving involves transferring the records from their primary storage location to a secondary storage location, where they can be accessed and retrieved when needed. Archiving moved data from live systems to free up system resources.</td></tr>
<tr><td>Q5(a)(i)</td><td>
</td></tr>
<tr><td>Q5 (a)(ii)</td><td>1. Each node has at most 2 children: a left child and a right child.
2. For every node in the tree, the left subtree contains only nodes with smaller values than the node and the right subtree contains only nodes with greater values than the node.</td></tr>
<tr><td>Q5 (a) (iii)</td><td>Madrid will be inserted to the left of New York.
Changes:
bst[6].dataValue ← Madrid
bst[6].leftPtr ← -1
bst[2].leftPtr ← 6
freePtr ← 7</td></tr>
<tr><td>Q5 (b)</td><td>1. A static data structure has a fixed size that cannot be changed during runtime. If the amount of data exceeds the initial allocation, it cannot dynamically expand, leading to limitations in handling larger datasets.
2. If a static data structure is initialized with more memory than needed, the unused space remains allocated and cannot be repurposed, resulting in inefficient memory utilization.</td></tr>
<tr><td>Q5 (c)(i)</td><td><pre>PROCEDURE reverseInOrder(index: INTEGER)
    IF index <> -1
       reverseInOrder(bst[index].rightPtr
       OUPUT bst[index].dataValue
       reverseInOrder(bst[index].leftPtr)
    ENDIF
ENDPROCEDURE</pre></td></tr>
</table>

	`// When called, the procedure should call with` `//reverseInOrder(rootPtr)`
Q5 (c)(ii)	Tokyo, Rome, Paris, New York, *Madrid, London, Beijing *bst with Madrid added
Q6 (a)	Client-server architecture is a network model that has one (or more) device acting as a server, while other devices are clients that request specific services from a server and the servers respond with the required data for the given request.
Q6 (b)	An **intranet** is a private, internal network within an organization that uses internet protocols to share information, resources, and services securely among its members, typically accessible only to authorized users.
Q6 (c)	Networking protocols are required to ensure standardized communication between devices in a network. They define the rules and conventions for data transmission, error handling, addressing, and routing, allowing devices from different manufacturers or with different operating systems to communicate effectively and securely.
Q6 (d)(i)	The purpose of a digital signature is to verify the authenticity, integrity, and non-repudiation of a digital message or document. It ensures that the sender is genuine (authenticity), the message has not been altered during transmission (integrity), and the sender cannot deny having sent the message (non-repudiation).
Q6 (d) (ii)	**Document Hashing:** Alice first creates a digest of the document she wants to send. A hashing algorithm is used for this purpose. The hash represents the contents of the document in a condensed form. **Signing the Hash:** Alice uses her private key to encrypt the hash of the document. This encrypted hash is the digital signature. The digital signature ensures that the document's content cannot be tampered with without invalidating the signature. **Attaching the Signature to the Document:** Alice attaches the digital signature to the original document as part of the email and send them to Bob via email.
Q6 (e)	**Verifying the Signature:** Bob receives the email with the signed document and the attached digital signature. Bob uses Alice's public key (which is either shared in advance) to decrypt the digital signature to obtain the original digest that Alice created for the document. **Rehashing the Document:** Bob independently hashes the received document using the same hashing algorithm Alice used. **Comparing Hashes:** Bob compares the digest he generated with the digest decrypted from the digital signature. If they match, it proves that the document has not been altered and that the signature is valid. If the hashes don't match, it means the document has been tampered with, or the signature is invalid. Alice's digital signature ensures the authenticity, integrity, and non-repudiation of the document she is sending to Bob.

H2 Computing P1 Ans Key